I- Introduction

Definition

If $f(x)$ is as close as we like to some real number A for all x sufficiently close to (but not equal to) a, then we say that $f(x)$ has a limit of A as x approaches a, and we write $\lim_{x\to a} f(x) = A$.

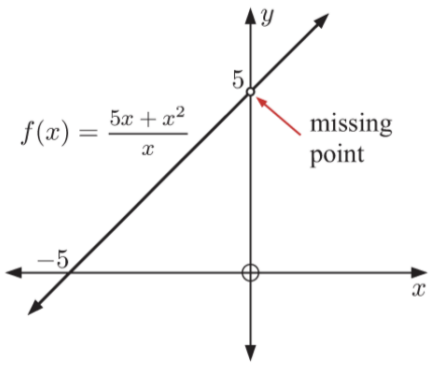
For example, suppose we wish to find the limit of $f(x) = \dfrac{5x + x^2}{x}$ as $x \to 0$.

It is tempting for us to simply substitute $x = 0$ into $f(x)$. However, in doing this, not only do we get the meaningless value of $\frac{0}{0}$, but also we ignore the basic limit definition.

Instead, observe that if $f(x) = \dfrac{5x + x^2}{x} = \dfrac{x(5 + x)}{x}$

$$\text{then} \quad f(x) = \begin{cases} 5 + x & \text{if } x \neq 0 \\ \text{is undefined if } x = 0. \end{cases}$$

The graph of $y = f(x)$ is therefore the straight line $y = x+5$ with a "hole" at $(0, 5)$. The "hole" is a **point of discontinuity** of the function.



However, even though this point is missing, the *limit* of $f(x)$ as x approaches 0 does exist. In particular, as $x \to 0$ from either direction, $f(x) \to 5$.

We write $\lim\limits_{x\to 0} \dfrac{5x + x^2}{x} = 5$ which reads:

"the limit as x approaches 0, of $\dfrac{5x + x^2}{x}$, is 5".

Property

Consider functions $f(x)$ and $g(x)$ for which $\lim_{x\to a} f(x) = l$ and $\lim_{x\to a} g(x) = m$, where a, l, $m \in \mathbb{R}$.

- $\lim_{x\to a}(f(x) \pm g(x)) = l \pm m$
- $\lim_{x\to a} f(x)\, g(x) = lm$
- $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{l}{m}$ provided $m \neq 0$

Example

Evaluate :

1. $\lim_{x\to 2} x^2$
2. $\lim_{x\to 0} \dfrac{x^2 + 3x}{x}$
3. $\lim_{x\to 3} \dfrac{x^2 - 9}{x - 3}$

EXERCISE 1

Evaluate:

a $\lim\limits_{x\to 1} \dfrac{x^2-3x}{x}$ b $\lim\limits_{h\to 2} \dfrac{h^2+5h}{h}$ c $\lim\limits_{x\to 0} \dfrac{x-1}{x+1}$

EXERCISE 2

Evaluate:

a $\lim\limits_{x\to 3} (x+4)$ b $\lim\limits_{x\to -1} (5-2x)$ c $\lim\limits_{x\to 4} (3x-1)$

d $\lim\limits_{x\to 2} (5x^2-3x+2)$ e $\lim\limits_{h\to 0} h^2(1-h)$ f $\lim\limits_{x\to 0} (x^2+5)$

EXERCISE 3

Explain why $\lim\limits_{x\to 0} \dfrac{x}{x} = 1$, even though $\dfrac{0}{0}$ is meaningless.

EXERCISE 4

Evaluate:

a $\lim\limits_{x\to 0} \dfrac{x^2-3x}{x}$ b $\lim\limits_{x\to 0} \dfrac{x^2+5x}{x}$ c $\lim\limits_{x\to 0} \dfrac{2x^2-x}{x}$

d $\lim\limits_{h\to 0} \dfrac{2h^2+6h}{h}$ e $\lim\limits_{h\to 0} \dfrac{3h^2-4h}{h}$ f $\lim\limits_{h\to 0} \dfrac{h^3-8h}{h}$

g $\lim\limits_{x\to 1} \dfrac{x^2-x}{x-1}$ h $\lim\limits_{x\to 2} \dfrac{x^2-2x}{x-2}$ i $\lim\limits_{x\to 3} \dfrac{x^2-x-6}{x-3}$

So far we have considered cases where the function $f(x)$ approaches some finite value A as x approaches a from either side. More formally, we can write this using the one-sided limits $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$.

Definition

The limit $\lim_{x\to a} f(x)$ exists and is equal to the finite value A if $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = A$. We say that $f(x)$ converges to A as x approaches a.

Remark. *We cannot write* $\lim_{x\to 0} f(x) = \infty$ *because* ∞ *is not a number. Rather,* $\lim_{x\to 0} f(x)$ *does not exist.*

Example

Find, if possible :

1. $\lim_{x\to 1} \dfrac{1}{x}$
2. $\lim_{x\to 0} \dfrac{1}{x}$

EXERCISE 5

a Sketch the graph of $y = -\dfrac{1}{(x-3)^2}$.

b Find, if possible:

i $\lim\limits_{x\to 0}\left(-\dfrac{1}{(x-3)^2}\right)$ ii $\lim\limits_{x\to 3}\left(-\dfrac{1}{(x-3)^2}\right)$ iii $\lim\limits_{x\to 5}\left(-\dfrac{1}{(x-3)^2}\right)$

EXERCISE 6

Find, if possible: a $\lim\limits_{x\to -1}(3x+2)$ b $\lim\limits_{x\to -2}\dfrac{x^2+x-2}{x-2}$

EXERCISE 7

Consider the function $f(x) = \dfrac{x}{x^2-2x}$.

a For what values of x is $f(x)$ undefined?

b Sketch the graph of $y = f(x)$.

c Find, if possible: i $\lim\limits_{x\to 2} f(x)$ ii $\lim\limits_{x\to 1} f(x)$ iii $\lim\limits_{x\to 0} f(x)$

EXERCISE 8

Let $f(x) = \begin{cases} x^2, & x > 3 \\ 5, & x = 3 \\ 3x, & x < 3. \end{cases}$

a Sketch the graph of $y = f(x)$.

b Evaluate $\lim\limits_{x\to 3^-} f(x)$ and $\lim\limits_{x\to 3^+} f(x)$.

c Does $\lim\limits_{x\to 3} f(x)$ exist? Explain your answer.

EXERCISE 9

Given $f(x) = \begin{cases} \sin x, & x < 0 \\ \dfrac{x^2-3x+5}{5}, & x > 0 \end{cases}$, discuss:

a $\lim\limits_{x\to 0^-} f(x)$ b $\lim\limits_{x\to 0^+} f(x)$ c $\lim\limits_{x\to 0} f(x)$

EXERCISE 10

Find, if possible:

a $\lim\limits_{x\to 0}\dfrac{\sin x}{e^x}$ b $\lim\limits_{x\to \pi}\dfrac{\sin x}{1-\cos x}$ c $\lim\limits_{\theta\to 0}\dfrac{\cos\theta}{\theta}$ d $\lim\limits_{x\to 2}\dfrac{\ln x}{\sqrt{2-x}}$

EXERCISE 11

Let $f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{otherwise.} \end{cases}$

a Find: i $f(\frac{1}{2})$ ii $f(\sqrt{2})$ iii $f(0)$

b Find, if possible: i $\lim\limits_{x\to \frac{1}{2}} f(x)$ ii $\lim\limits_{x\to \sqrt{2}} f(x)$ iii $\lim\limits_{x\to 0} f(x)$

Explain your answers.

II- Specific limits

a) Infinity

Property

$$\lim_{x\to\infty} \frac{1}{x^n} = 0 \quad \text{for all } n > 0.$$

$$\lim_{x\to-\infty} \frac{1}{x^n} = 0 \quad \text{for all } n \in \mathbb{Z}^+.$$

Proof

Notice that as $x \to \infty$, $1 < x < x^2 < x^3 < \ldots$ and as x gets very large, the value of $\frac{1}{x}$ gets very small. In fact, we can make $\frac{1}{x}$ as close to 0 as we like by making x large enough. This means that $\lim_{x\to\infty} \frac{1}{x} = 0$, even though $\frac{1}{x}$ never actually reaches 0.

Example

Evaluate :

1. $\lim_{x\to\infty} \frac{2x+3}{x-4}$
2. $\lim_{x\to\infty} \frac{x^2-3x+2}{1-x^2}$
3. $\lim_{x\to\infty} \frac{x^2+x+1}{x-2}$

EXERCISE 12

Evaluate:

a $\lim_{x\to\infty} \frac{3x-2}{x+1}$ b $\lim_{x\to\infty} \frac{1-2x}{3x+2}$ c $\lim_{x\to\infty} \frac{x}{1-x}$

d $\lim_{x\to\infty} \frac{x^2+3}{x^2-1}$ e $\lim_{x\to\infty} \frac{x^2-2x+4}{4x^2-x-1}$ f $\lim_{x\to\infty} \frac{x+4}{2x^2+x-1}$

g $\lim_{x\to\infty} \frac{x+5}{x^2-x+4}$ h $\lim_{x\to\infty} \frac{x^2+3x-4}{x-1}$ i $\lim_{x\to\infty} \frac{2x^2-x+1}{5x+3}$

Example

1. Discuss the behaviour of $f(x) = \frac{2-x}{1+x}$ near its asymptotes, and hence deduce their equations.
2. If they exist, state the values of $\lim_{x\to-\infty} f(x)$ and $\lim_{x\to\infty} f(x)$.

EXERCISE 13

For each of the following functions:

i Discuss the behaviour near its asymptotes, and hence deduce their equations.

ii If they exist, state the values of $\lim_{x\to-\infty} f(x)$ and $\lim_{x\to\infty} f(x)$.

a $f(x) = \frac{1}{x}$ **b** $f(x) = \frac{3x-2}{x+3}$ **c** $f(x) = \frac{1-2x}{3x+2}$ **d** $f(x) = \frac{x}{1-x}$

Example

Find, if possible :

1. $\lim_{x\to-\infty}(3-e^{-x})$
2. $\lim_{x\to\infty}(3-e^{-x})$

EXERCISE 14

Find, if possible:

a $\lim_{x\to-\infty}(2e^{-x}-3)$ and $\lim_{x\to\infty}(2e^{-x}-3)$

b $\lim_{x\to-\infty}(e^{-x}+5e^{\frac{1}{x}})$ and $\lim_{x\to\infty}(e^{-x}+5e^{\frac{1}{x}})$

b) Trigonometry

Property

If θ is in radians, then $\lim_{\theta\to0}\frac{\sin\theta}{\theta} = 1$.

Example

Find $\lim_{\theta\to0}\frac{\sin 3\theta}{\theta}$.

EXERCISE 15

Find:

a $\lim_{\theta\to0}\frac{\sin 2\theta}{\theta}$ **b** $\lim_{\theta\to0}\frac{\theta}{\sin\theta}$ **c** $\lim_{\theta\to0}\frac{\tan\theta}{\theta}$

d $\lim_{\theta\to0}\frac{\sin^2\theta}{\theta}$ **e** $\lim_{\theta\to0}\frac{\sin\theta\sin 4\theta}{\theta^2}$ **f** $\lim_{h\to0}\frac{\sin\frac{h}{2}\cos h}{h}$

g $\lim_{n\to\infty} n\sin\frac{3\pi}{n}$ **h** $\lim_{x\to0}\frac{\sin 7x}{4x}$ **i** $\lim_{x\to0}\frac{x^2+x}{\sin 2x}$

EXERCISE 16

Evaluate, where possible:

a $\lim_{x\to 0} \dfrac{x+\sin x}{x-\sin x}$ by multiplying by $\dfrac{\frac{1}{x}}{\frac{1}{x}}$

b $\lim_{h\to 0} \dfrac{\cos h - 1}{h}$ by multiplying by $\dfrac{\cos h + 1}{\cos h + 1}$

c $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$ by multiplying by $\dfrac{1+\cos x}{1+\cos x}$.

III- Continuity

Definition

A function f is continuous at $\boldsymbol{x = a}$ if $f(a)$ and $\lim_{x\to a} f(x)$ exist and are equal.

f is also continuous at $x = a$ if :

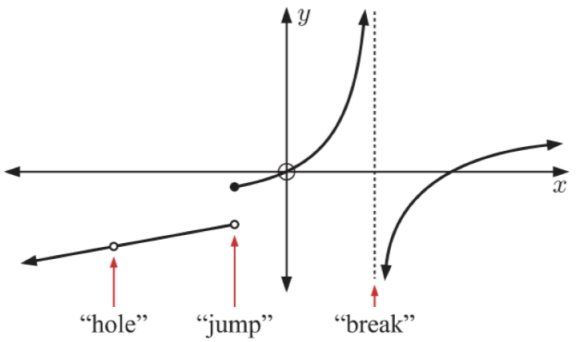
— a is the left endpoint of an interval on which f is defined, and $f(a)$ and $\lim_{x\to a^+} f(x)$ exist and are equal, or

— a is the right endpoint of an interval on which f is defined, and $f(a)$ and $\lim_{x\to a^-} f(x)$ exist and are equal.

Example

If f is *not* continuous at $x = a$, we say that f is *discontinuous* at $x = a$ or *has a discontinuity* at $x = a$.

Graphically, we have seen that isolated points of discontinuity are indicated on the graph of $y = f(x)$ by a "hole", a "jump", or a "break".



If f has a "hole" at $x = a$, then $\lim_{x\to a} f(x)$ exists. The "hole" is a **removable discontinuity** which can be removed by defining a new function based on f but which takes the value $\lim_{x\to a} f(x)$ when $x = a$.

Discontinuities due to "jumps" and "breaks" cannot be removed by simply redefining the value of the function there. They are called **essential discontinuities**.

Example

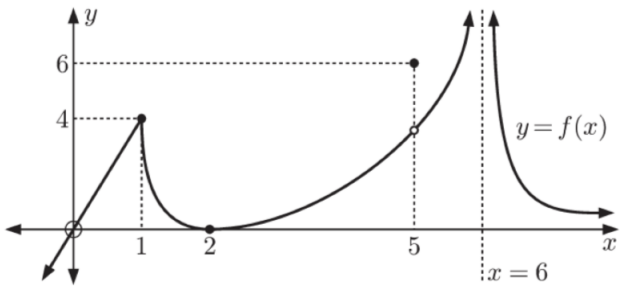
Discuss the continuity of the following functions. If there is a removable discontinuity, describe how it could be removed.

1. $f(x) = \dfrac{1}{x-5}$
2. $f(x) = \begin{cases} 1, & x \geq 0 \\ -1, & x < 0 \end{cases}$
3. $f(x) = \begin{cases} x^2, & x \neq 2 \\ 6, & x = 2 \end{cases}$

EXERCISE 17

The graph of $y = f(x)$ is shown alongside.
State the value(s) of x for which f:

a has an essential discontinuity
b has a removable discontinuity
c is continuous.



EXERCISE 18

Discuss the continuity of the following functions. If there is a removable discontinuity, describe how it could be removed.

a $f(x) = \dfrac{3}{x+2}$ b $f(x) = \dfrac{x^2+x}{x+1}$ c $f(x) = \dfrac{x-4}{x^2-2x-8}$

d $f(x) = \ln(x^2)$ e $f(x) = \begin{cases} x^2+2, & x \geqslant 3 \\ 5, & x < 3 \end{cases}$ f $f(x) = \tan x \cos x$

EXERCISE 19

Find, if possible, the value(s) of $k \in \mathbb{R}$ for which f is continuous on $\mathbb{R}$. Explain your answers.

a $f(x) = \begin{cases} \dfrac{x^3-1}{x-1}, & x \neq 1 \\ k, & x = 1 \end{cases}$ b $f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ c $f(x) = \begin{cases} k+1, & x < 2 \\ x^2, & x \geqslant 2 \end{cases}$

d $f(x) = \begin{cases} \dfrac{1}{x}, & x > 0 \\ k, & x \leqslant 0 \end{cases}$ e $f(x) = \begin{cases} kx, & x > 0 \\ 0, & x \leqslant 0 \end{cases}$ f $f(x) = \begin{cases} k+x, & x \geqslant 2 \\ \lvert k+2 \rvert, & x < 2 \end{cases}$

EXERCISE 20

An alternative definition of continuity is:
"f is continuous at $x = a$ if, for any sequence $\{x_n\}$ such that $\lim_{n\to\infty} x_n = a$, then $\lim_{n\to\infty} f(x_n) = f(a)$."

The **Dirichlet function** is defined by $g(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$.

a Consider the sequence $x_n = a + \frac{\sqrt{2}}{n}$, $a \in \mathbb{Q}$, $n \in \mathbb{Z}^+$. Show that $g(x)$ is discontinuous at $x = a$.

b Now suppose $a \notin \mathbb{Q}$, and that x_n is the decimal expansion of a to n decimal places, $n \in \mathbb{Z}^+$. Show that $g(x)$ is discontinuous at $x = a$.

c Hence discuss the continuity of the Dirichlet function.

IV- L'Hôpital's rule

Example

- The curve $f(x) = \frac{\sin x}{x}$ has a discontinuity at $x = 0$.
  In the limit $\lim_{x\to 0} \frac{\sin x}{x}$, both $\lim_{x\to 0} \sin x = 0$ and $\lim_{x\to 0} x = 0$.
  We therefore need to consider something of the form $\frac{0}{0}$.
  In this particular case, we have seen previously that $\lim_{x\to 0} \frac{\sin x}{x} = 1$.
- For the curve $f(x) = x^2 e^{-x}$, consider the behaviour as $x \to \infty$.
  We know that x^2 becomes infinitely large, and that $\lim_{x\to\infty} e^{-x} = 0$.
  We therefore need to consider something of the form $\infty \times 0$.

There are three **indeterminate forms** which we need to deal with :

Type	*Description*
$\frac{0}{0}$	$\lim_{x\to a} \frac{f(x)}{g(x)}$ where $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$
$\frac{\infty}{\infty}$	$\lim_{x\to a} \frac{f(x)}{g(x)}$ where $f(x) \to \pm\infty$ and $g(x) \to \pm\infty$ as $x \to a$
$0 \times \infty$	$\lim_{x\to a}[f(x)\,g(x)]$ where $\lim_{x\to a} f(x) = 0$ and $g(x) \to \pm\infty$ as $x \to a$

Theorem

To address these types of limits, we use **l'Hôpital's rule** :

Suppose $f(x)$ and $g(x)$ are differentiable and $g'(x) \neq 0$ on an interval that contains the point $x = a$. If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$, or, if as $x \to a$, $f(x) \to \pm\infty$ and $g(x) \to \pm\infty$, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

provided the limit on the right exists.

Proof

Proof for the case of l'Hôpital's rule where $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, $f'(x)$ and $g'(x)$ continuous, $g'(a) \neq 0$: Since $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ where f and g are continuous, $f(a) = g(a) = 0$

$$\therefore \quad \lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f(x) - f(a)}{g(x) - g(a)}$$

$$= \lim_{x\to a} \frac{\dfrac{f(x) - f(a)}{x - a}}{\dfrac{g(x) - g(a)}{x - a}} \qquad \{\text{since } x \neq a\}$$

$$= \frac{\lim_{x\to a} \dfrac{f(x) - f(a)}{x - a}}{\lim_{x\to a} \dfrac{g(x) - g(a)}{x - a}} \qquad \{\text{limit laws}\}$$

$$= \frac{f'(a)}{g'(a)}$$

$$= \frac{\lim_{x\to a} f'(x)}{\lim_{x\to a} g'(x)} \qquad \{\text{since } f'(x) \text{ and } g'(x) \text{ are continuous}\}$$

$$= \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

Example

Use l'Hôpital's rule to evaluate :

1. $\lim_{x\to 0} \dfrac{2^x - 1}{x}$
2. $\lim_{x\to\infty} xe^{-x}$
3. $\lim_{x\to\infty} \dfrac{\ln x}{x}$

EXERCISE 21

Evaluate, if possible, using l'Hôpital's rule.

a $\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2}$ **b** $\lim\limits_{x\to 0} \dfrac{e^x-1-x}{x^2}$ **c** $\lim\limits_{x\to 1} \dfrac{\ln x}{x-1}$

d $\lim\limits_{x\to \infty} \dfrac{e^x}{x}$ **e** $\lim\limits_{x\to 0^+} x\ln x$ **f** $\lim\limits_{x\to 0} \dfrac{\arctan x}{x}$

g $\lim\limits_{x\to 0} \dfrac{x^2+x}{\sin 2x}$ **h** $\lim\limits_{x\to 0^+} \dfrac{\sin x}{\sqrt{x}}$ **i** $\lim\limits_{x\to 0} \dfrac{x+\sin x}{x-\sin x}$

j $\lim\limits_{x\to 0^+} x^2\ln x$ **k** $\lim\limits_{x\to 0} \dfrac{a^x-b^x}{\sin x}$, where $a, b>0$ are real constants.

EXERCISE 22

Attempt to find $\lim\limits_{x\to \frac{\pi}{2}^-} \dfrac{\tan x}{\sec x}$ using l'Hôpital's rule.

EXERCISE 23

Consider the function $f(x)=\dfrac{e^x-1}{x^2+x}$.

a State the domain of f. **b** Find $f'(x)$ and state its domain.

c Find the turning point of $y=f(x)$.

d Discuss the behaviour of $f(x)$: **i** as $x\to-\infty$ **ii** near $x=0$.

e Sketch $y=f(x)$, showing the features you have found.

Example

Find $\lim_{x\to 0^+} \dfrac{\ln(\cos 3x)}{\ln(\cos 2x)}$.

EXERCISE 24

Evaluate, if possible:

a $\lim\limits_{x\to 0^+} \dfrac{\ln(\cos 5x)}{\ln(\cos 3x)}$ **b** $\lim\limits_{x\to \frac{\pi}{2}^-} \dfrac{\ln(\sin 2x)}{\ln(\sin 3x)}$

Example

Evaluate $\lim_{x\to \frac{\pi}{2}^-}(\sec x-\tan x)$.

EXERCISE 25

Evaluate $\lim\limits_{x\to 0^+}\left(\dfrac{1}{x}-\dfrac{1}{\sin x}\right)$ using l'Hôpital's rule *twice*.

Example

Evaluate, if possible : $\lim_{x\to\infty} \dfrac{e^x}{x^n}$ where $n\in\mathbb{Z}^+$.

EXERCISE 26

a Prove that $\lim\limits_{x\to\infty} x\ln\left(1+\frac{1}{x}\right)=1$.

b By writing $\left(1+\frac{1}{x}\right)^x = e^{x\ln\left(1+\frac{1}{x}\right)}$ and using the fact that $f(x)=e^x$ is continuous on $\mathbb{R}$, prove that $\lim\limits_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e$.

c Prove that for $a\neq 0$, $\lim\limits_{x\to\infty}\left(1+\frac{a}{x}\right)^x = e^a$.